

AYE
TECH HUB

• HVAC ENGINEERING • LESSON 09

Psychrometrics

The Science of Moist Air in HVAC Design

Lesson 09 of 30 | HVAC Engineering Programme | AYE Tech Hub

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- What psychrometrics is and why every HVAC engineer must master it
- 7 key air properties: DBT, WBT, dew point, RH, humidity ratio, enthalpy, SHR
- How to read and use the psychrometric chart step by step
- Cooling, dehumidification, heating and humidification processes plotted
- Real cooling coil selection example using psychrometric calculations
- Worked calculations with ASHRAE standard conditions

What Is Psychrometrics and Why Does It Matter?

Psychrometrics is the branch of engineering thermodynamics that studies the properties of mixtures of dry air and water vapour. In HVAC engineering, it is the mathematical foundation for every system that conditions air — cooling, heating, humidifying, or dehumidifying. Without psychrometrics, you cannot size a cooling coil, calculate the capacity of a chiller, select a dehumidifier, or correctly interpret a weather data file. It is not optional knowledge for an HVAC engineer — it is the core.

The air inside and around any building is never purely dry air. It always contains water vapour — a mixture that behaves differently at different temperatures and relative humidities. The **psychrometric chart** is the graphical tool that maps all these properties simultaneously, allowing engineers to read off or calculate any air condition and visualise what happens when that air is processed through HVAC equipment.

The 7 Key Psychrometric Properties

Property	Symbol	Unit	Definition	Typical Range (HVAC)
Dry Bulb Temperature	DBT (T)	°C	Air temperature measured by an ordinary thermometer — does not account for moisture effects	-10°C to +50°C
Wet Bulb Temperature	WBT (T*)	°C	Temperature of air cooled to saturation by evaporating water into it. Always ≤ DBT. Equals DBT at 100% RH	DBT – 15°C to DBT
Dew Point Temperature	DPT (Td)	°C	Temperature at which air must be cooled (at constant pressure) to become saturated. Below this, condensation occurs	Varies with humidity
Relative Humidity	RH (φ)	%	Ratio of actual water vapour pressure to saturation pressure at the same temperature	0% to 100%
Humidity Ratio	W (ω)	g/kg or kg/kg	Mass of water vapour per unit mass of dry air. Also called specific humidity or moisture content	0 to 30 g/kg
Enthalpy	h	kJ/kg	Total heat content of moist air per kg of dry air: sum of sensible heat (temperature) and latent heat (moisture)	0 to 120 kJ/kg
Specific Volume	v	m³/kg	Volume occupied by 1 kg of dry air plus its associated water vapour. Inverse of air density	0.77 to 1.0 m³/kg

Relationships Between the Properties

These seven properties are not independent — if you know any two of them for a given air sample, you can calculate all the others. This is why the psychrometric chart is so powerful: plot one point (defined by any two properties) and read off the rest. In practice, **DBT and RH** are the easiest to measure in the field and are the most common pair used to define an air condition.

Formula	Description	When to Use
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$W = 0.622 \times P_v / (P - P_v)$	Humidity ratio from vapour pressure P_v and total pressure P	Detailed psychrometric calculations
$h = 1.006 \times T + W \times (2501 + 1.86 \times T)$	Enthalpy (kJ/kg dry air) from DBT T (°C) and humidity ratio W (kg/kg)	Cooling coil load, chiller sizing
$RH = P_v / P_{vs} \times 100\%$	Relative humidity from actual and saturation vapour pressures	Comfort and condensation assessment
$P_{vs} = \exp(16.62 - 4030/(T + 235))$	Saturation vapour pressure (simplified Antoine equation)	Quick dew point and WBT estimation
$Q_{\text{sensible}} = \dot{m} \times 1.006 \times \Delta T$	Sensible cooling/heating load (kW) from mass flow and temp change	AHU coil sizing, fan coil selection
$Q_{\text{latent}} = \dot{m} \times 2501 \times \Delta W$	Latent load (kW) from mass flow and humidity ratio change	Dehumidifier sizing, condensate drainage
$SHR = Q_{\text{sensible}} / Q_{\text{total}}$	Sensible Heat Ratio — fraction of total load that is sensible	Selecting AC equipment for specific space type

How to Read the Psychrometric Chart

The psychrometric chart is one of the most information-dense engineering diagrams in existence. Once you can read it confidently, complex HVAC calculations that would otherwise require multiple equations become straightforward graphical lookups. Here is how to navigate it systematically:

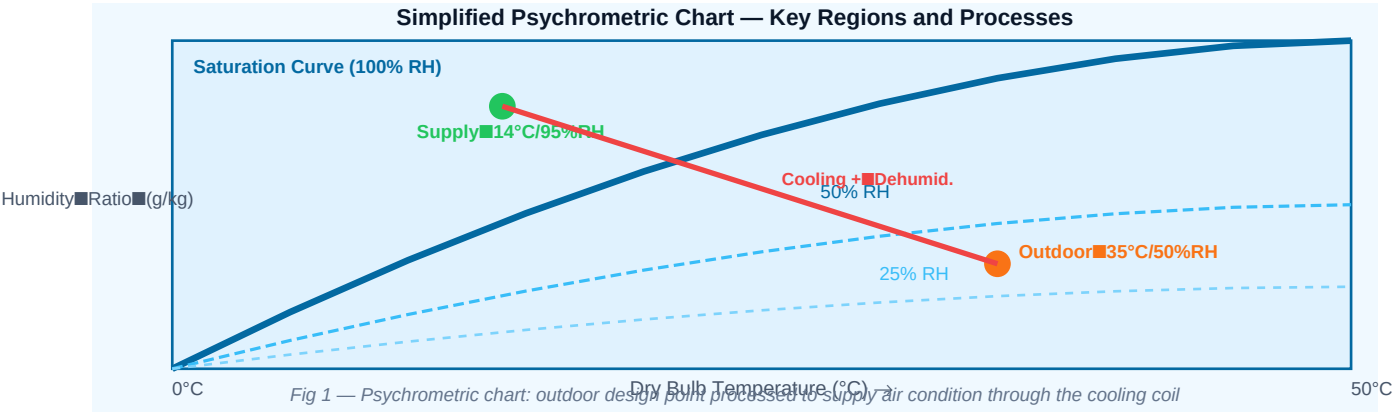


Chart Element	Location	What It Shows
Dry Bulb Temperature lines	Vertical lines (x-axis)	Temperature of the air, read at the bottom axis. Range typically -10°C to +55°C
Humidity Ratio lines	Horizontal lines (y-axis)	Water vapour content in g/kg dry air, read on the right axis
Saturation curve	Curved line on the left	100% relative humidity — air is fully saturated; cannot hold more moisture at that temperature
Relative humidity curves	Curved lines at 10%, 20%...90%	Constant RH lines fanning out from the lower left toward upper right

Wet bulb temperature lines	Diagonal lines sloping left-down	Constant WBT — used to find enthalpy and locate the dew point
Enthalpy lines	Parallel to WBT lines (extended left)	Total heat content in kJ/kg dry air — read on the left/top scale
Specific volume lines	Steep diagonal lines	Air density inverse in m³/kg — used for fan and duct airflow calculations

Step-by-Step: Locating an Air Condition on the Chart

Example: Outdoor air at 32°C DBT and 60% RH. Find all other properties.

Step	Action	Result for 32°C / 60% RH
1	Find 32°C on the x-axis (bottom). Draw a vertical line upward	Vertical line at 32°C
2	Find the 60% RH curve. Follow it from the saturation curve outward	Curved line at 60%
3	Mark the intersection — this is your air condition point	Point P defined
4	Read humidity ratio (W) on the right axis at the height of point P	W ≈ 18.8 g/kg
5	Follow the nearest wet-bulb temperature line to the saturation curve to read WBT	WBT ≈ 25.2°C
6	Read enthalpy on the left/top scale along the WBT line	h ≈ 80.4 kJ/kg
7	Read dew point: follow horizontal line from P to saturation curve, read x-axis	DPT ≈ 23.6°C

PRACTICAL NOTE — Moisture condenses on any surface whose temperature is below the dew point of the surrounding air. In the example above, any surface below 23.6°C will attract condensation. This is why chilled water pipes below the dew point must be insulated, and why cold windows "sweat" in humid climates.

The Four Fundamental Psychrometric Processes

Every HVAC air-handling process can be represented as a line or path on the psychrometric chart — moving from the entering air condition to the leaving air condition. Understanding these four fundamental processes allows you to design, select, and troubleshoot any air conditioning system.

Four Fundamental Psychrometric Processes in HVAC Systems



Fig 2 — DBT=Dry Bulb Temp | W=Humidity Ratio | RH=Relative Humidity

Process 1: Sensible Cooling (Horizontal Line — Left)

The air temperature (DBT) decreases while the humidity ratio (W) remains constant. No moisture is added or removed — only sensible heat is extracted. On the chart, this appears as a horizontal line moving left. This process occurs in the **pre-cooling section of an AHU before the dew point is reached**, or in a sensible-only cooling device (such as a direct evaporative cooler in some conditions).

Load formula: $Q_{\text{sensible}} = \dot{V} \times 1.006 \times (T_{\text{in}} - T_{\text{out}}) \text{ kW}$ Example: 2 kg/s of air cooled from 35°C to 24°C
 $= 2 \times 1.006 \times 11 = 22.1 \text{ kW}$

Process 2: Cooling and Dehumidification (Curve — Down-Left)

This is the most common process in air conditioning. The air temperature **and** humidity ratio both decrease. Moisture condenses out on the cooling coil surface (which is below the dew point temperature of the air). The air leaves the coil approximately saturated at the coil surface temperature, called the **Apparatus Dew Point (ADP)**. The further the leaving air condition is from the ADP, the higher the Bypass Factor (BF) of the coil — meaning more air bypasses the coil surface without being fully conditioned.

Total load: $Q_{\text{total}} = \dot{V} \times (h_{\text{in}} - h_{\text{out}}) \text{ kW}$ **Latent load:** $Q_{\text{latent}} = \dot{V} \times 2501 \times (W_{\text{in}} - W_{\text{out}}) \text{ kW}$
Condensate rate: $\dot{m}_{\text{condensate}} = \dot{V} \times (W_{\text{in}} - W_{\text{out}}) \text{ kg/s}$

Process 3: Sensible Heating (Horizontal Line — Right)

The air temperature increases while humidity ratio remains constant. Used in heating coils (hot water or electric), post-cooling reheat coils, and terminal reheat systems for humidity control. On the chart: horizontal line moving right. **Load formula:** $Q_{\text{heat}} = \dot{V} \times 1.006 \times (T_{\text{out}} - T_{\text{in}}) \text{ kW}$

Process 4: Humidification (Near-Vertical Line — Up)

Water vapour or steam is added to the air, increasing the humidity ratio. If steam is injected (isothermal humidification), DBT barely changes and the line is nearly vertical. If water is evaporated by spraying (adiabatic humidification), both DBT drops and W increases — the line follows a WBT line toward the saturation curve. Used in hospitals, data centres, and museums where humidity must be maintained precisely.

Worked Example: Cooling Coil Selection

A commercial office in Tigray requires air conditioning. The design conditions are:

Outdoor: **38°C DBT / 30% RH** (hot dry climate)

Indoor setpoint: **24°C DBT / 50% RH**

Ventilation: **500 L/s** of outdoor air mixed with **1,500 L/s** recirculated room air

Supply air target: **14°C DBT** (standard cooling coil leaving condition)

Determine: Mixed air condition, cooling coil load, and Sensible Heat Ratio.

S t e p	Calculation	Result
1 . O u t d o o r a i r p r o p e r t i e s (3 8 ° C / 3 0 % R H)	From psychrometric chart: W_outdoor ≈ 12.8 g/kg, h_outdoor ≈ 70.9 kJ/kg	W=12.8 g/kg, h=70.9 kJ/kg

2 . R o o m a i r p r o p e r t i e s (2 4 ° C / 5 0 % R H)	From psychrometric chart: W _{room} ≈ 9.3 g/kg, h _{room} ≈ 47.5 kJ/kg	W=9.3 g/kg, h=47.5 kJ/kg
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3 M i x e d a i r c o n d i t i o n (2 5 % o u t d o o r)	$W_{mix} = 0.25 \times 12.8 + 0.75 \times 9.3 = 10.175 \text{ g/kg}$ $h_{mix} = 0.25 \times 70.9 + 0.75 \times 47.5 = 53.35 \text{ kJ/kg}$	$W_{mix} = 10.2 \text{ g/kg}$, $h_{mix} = 53.4 \text{ kJ/kg}$
4 M i x e d a i r D B T	$T_{mix} = 0.25 \times 38 + 0.75 \times 24 = 9.5 + 18 = 27.5^\circ\text{C}$	$T_{mix} = 27.5^\circ\text{C DBT}$

5 . S u p p l y a i r p r o p e r t i e s (1 4 ° C , n e a r s a t u r a t i o n)	From psychrometric chart: $W_{\text{supply}} \approx 9.5 \text{ g/kg}$ (at ADP $\sim 11^\circ\text{C}$, $\text{BF}=0.15$), $h_{\text{supply}} \approx 37.7 \text{ kJ/kg}$	$W=9.5 \text{ g/kg}$, $h=37.7 \text{ kJ/kg}$
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6 Mass flow rate of supply air	Total airflow = 500+1500 = 2,000 L/s = 2.0 m³/s m³/kg = 2.35 kg/s	■ = 2.35 kg/s
7 Total cooling load	Q_total = ■ × (h_mix – h_supply) = 2.35 × (53.4 – 37.7)	Q_total = 36.9 kW

8 . S e n si bl e c o ol in g lo a d	$Q_{\text{sensible}} = \dot{V} \times 1.006 \times (T_{\text{mix}} - T_{\text{supply}}) = 2.35 \times 1.006 \times (27.5 - 14)$	$Q_{\text{sensible}} = 31.9 \text{ kW}$
9 . L a t e n t c o ol in g lo a d	$Q_{\text{latent}} = Q_{\text{total}} - Q_{\text{sensible}} = 36.9 - 31.9$	$Q_{\text{latent}} = 5.0 \text{ kW}$

10.Sensible Heat Ratio (SHR)

$$\text{SHR} = Q_{\text{sensible}} / Q_{\text{total}} = 31.9 / 36.9$$

$$\text{SHR} = 0.86$$

RESULT — Select a cooling coil with minimum 37 kW capacity, a SHR of 0.86 (high sensible, low latent — typical for hot-dry climates like Tigray), and a leaving air temperature of 14°C. The chiller plant serving this coil must provide sufficient chilled water flow at 6–8°C entering water temperature.

ASHRAE Standard Conditions and Comfort Zones

ASHRAE Standard 55 defines the thermal comfort zone — the range of air conditions within which at least 80% of building occupants are thermally satisfied. Plotting these conditions on the psychrometric chart produces the well-known "comfort envelope." HVAC systems must maintain conditions within this zone throughout the year. The following table summarises key design conditions:

Condition	DBT (°C)	RH (%)	W (g/kg)	Enthalpy (kJ/kg)	Application
ASHRAE Comfort — Summer	23–26	30–60%	5.5–12.0	35–55	Office buildings, shopping centres
ASHRAE Comfort — Winter	20–24	30–50%	4.0–9.5	28–44	Offices, residences with heating
Hospital — Operating Theatre	18–21	40–60%	5.1–9.8	30–46	Infection control, sterile environment
Data Centre (ASHRAE A1)	15–32	8–80% (max dew point 17°C)	1.5–12.5	18–60	Server rooms, UPS rooms
Museum / Archive	16–21	45–55%	5.8–8.5	31–42	Artefact and document preservation

Cold Storage (vegetables)	2–5	85–95%	4.5–6.2	14–21	Fresh produce refrigeration
Outdoor design — Tigray (summer)	38–42	15–35%	7.5–14.5	57–79	HVAC design basis for cooling load

Humidity Control — Why Relative Humidity Matters

RH Level	Effect on People	Effect on Building / Equipment
Below 25% RH	Dry eyes, skin, throat irritation, increased static electricity, higher virus transmission	Wood furniture cracks and shrinks; static discharge damages electronics; increased dust
25–30% RH	Borderline acceptable — marginal comfort	Some static issues; acceptable for most spaces
30–60% RH	ASHRAE comfort zone — optimal for human health, productivity, and comfort	Safe for electronics, materials, and occupants; mould unlikely
60–70% RH	Muggy sensation; increased fatigue and discomfort; condensation risk on cold surfaces	Mould begins to grow on organic surfaces; structural moisture risk increases
Above 70% RH	Unacceptable comfort; significant health risk from mould spores and dust mites	Rapid mould growth; corrosion of metals; risk of legionella in cooling water

Common Psychrometric Mistakes in HVAC Design

Mistake	Consequence	Correct Approach
Using DBT alone to assess comfort	Overestimates comfort in humid conditions — a 28°C/80%RH space feels much hotter than 28°C/40%RH	Always specify both DBT and RH. Use enthalpy or apparent temperature for comfort assessment
Ignoring latent load in humid climates	System undersized for humidity — building stays cool but feels damp and muggy; mould risk	Calculate sensible AND latent load separately. Choose equipment SHR to match space SHR
Design based on mean temperature, not peak	System undersized for hottest design days — fails on the days it is needed most	Use 1% design dry bulb from ASHRAE Fundamentals for peak cooling; 99% for peak heating
Overlooking fresh air latent load	Calculations underestimate chiller capacity by 20–40% in humid climates	Calculate enthalpy difference (not just temperature difference) for outdoor air loads
Confusing WBT and DPT	Incorrect coil ADP selection leads to wrong bypass factor and coil geometry	WBT = evaporative saturation point. DPT = condensation point. They are different properties
Ignoring altitude correction	In highland areas (Tigray 2,000m+), air density is 18% lower — same volume flow carries less mass flow, less sensible capacity	Apply altitude correction factor: $p_{alt} = p_{sea} \times \exp(-0.0001255 \times \text{altitude})$

Key Takeaways — HVAC-009 Psychrometrics

- **Master two properties, derive seven:** Any two of DBT, WBT, RH, W, h, DPT, or v fully define the air condition. Learn to find all seven from any two.
- **The psychrometric chart is a calculator:** Plot entering and leaving conditions, draw the process line, and read loads from enthalpy and humidity ratio axes.
- **Sensible and latent loads are both real:** The latent load (dehumidification) is often the limiting factor in humid climates. Never ignore it.
- **SHR drives equipment selection:** Match equipment SHR to space SHR. Offices need SHR 0.80-0.90; humid residential spaces may need SHR 0.65-0.75.
- **The dew point controls condensation:** Any surface below the dew point will attract condensation. Insulate pipes, or raise surface temperature above dew point.
- **Altitude matters in highland Ethiopia:** At 2,000m (Tigray), air density is 18% lower. Apply altitude correction factors for all fan, duct and coil selections.
- **Next lesson - HVAC-010:** Load Calculation Methods - applying psychrometric principles to calculate peak heating and cooling loads for a complete building.